

Assignment 1

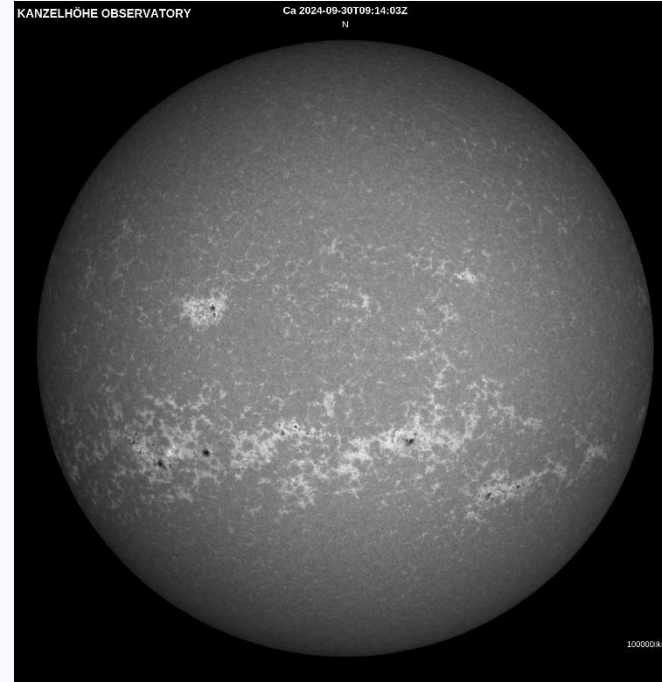
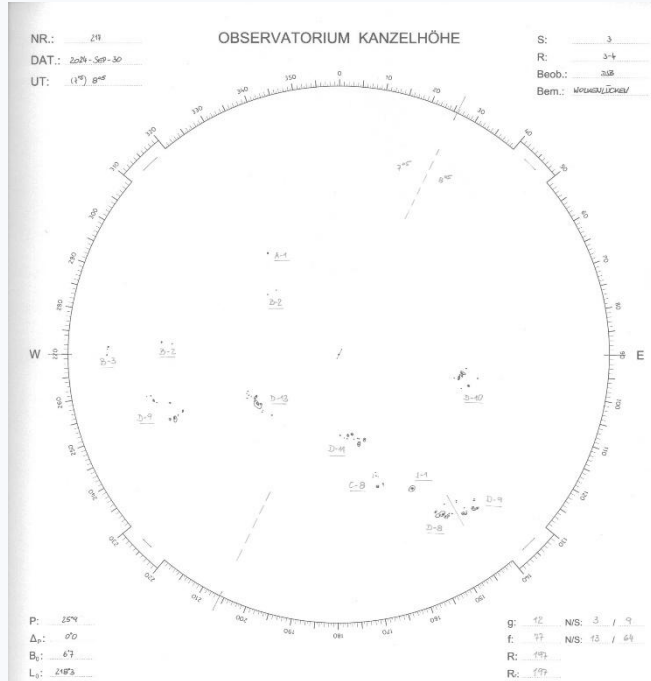
Experimental Data Processing

From noisy data to reliable predictions and decisions

Reconstructing the past:
Estimating solar radio flux F10.7 from sunspot records.

SUNSPOT NUMBER OBSERVATIONS

30 September 2024



$$R = k(n + 10g)$$

n – number of observed sunspots

g – number of observed sunspot groups

k – coefficient of a telescope

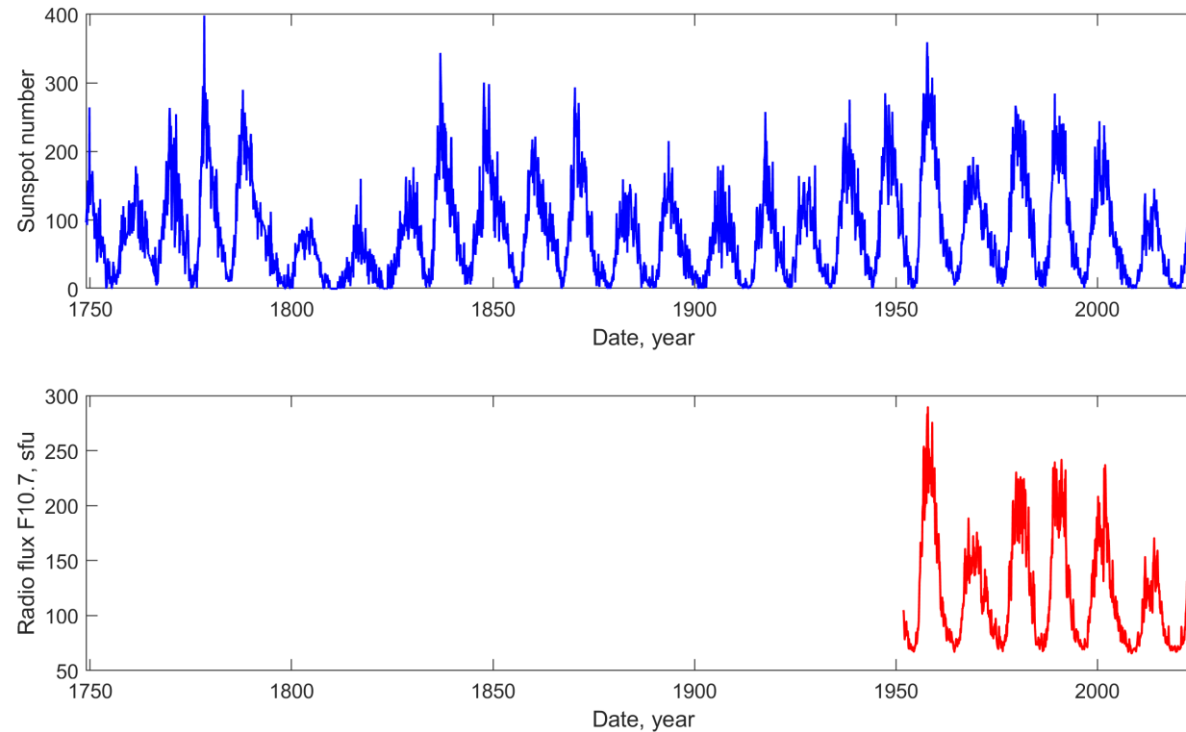
Around **80** cooperating stations
around the globe observe sunspot
numbers **daily**

Main indicators of solar activity

Sunspot number



From ground-based observatories



Solar radio flux at 10.7 cm (in sfu)

1 sfu = $10^{-22} \text{ W m}^{-2} \text{ Hz}$



Ottawa and
Pentinction
Observatory, Canada

$$R = k(n + 10g)$$

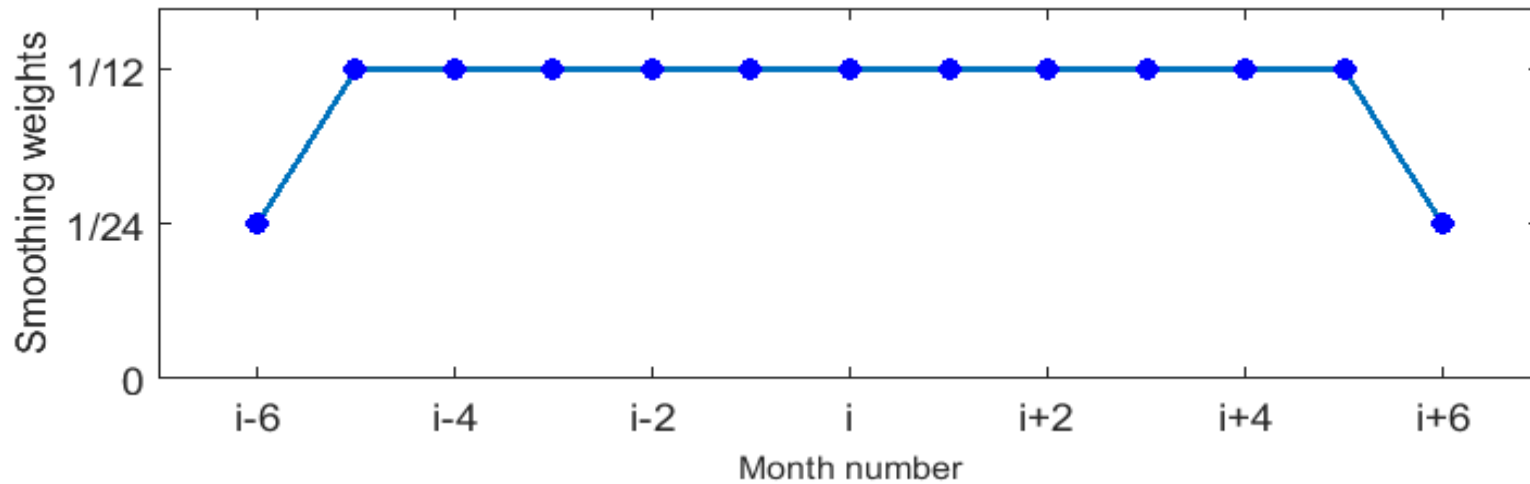
n – number of observed sunspots

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k – coefficient of a telescope

A measurement of radio
emission at a wavelength
of 10.7 cm (2800 MHz) from
all sources present on the solar disk

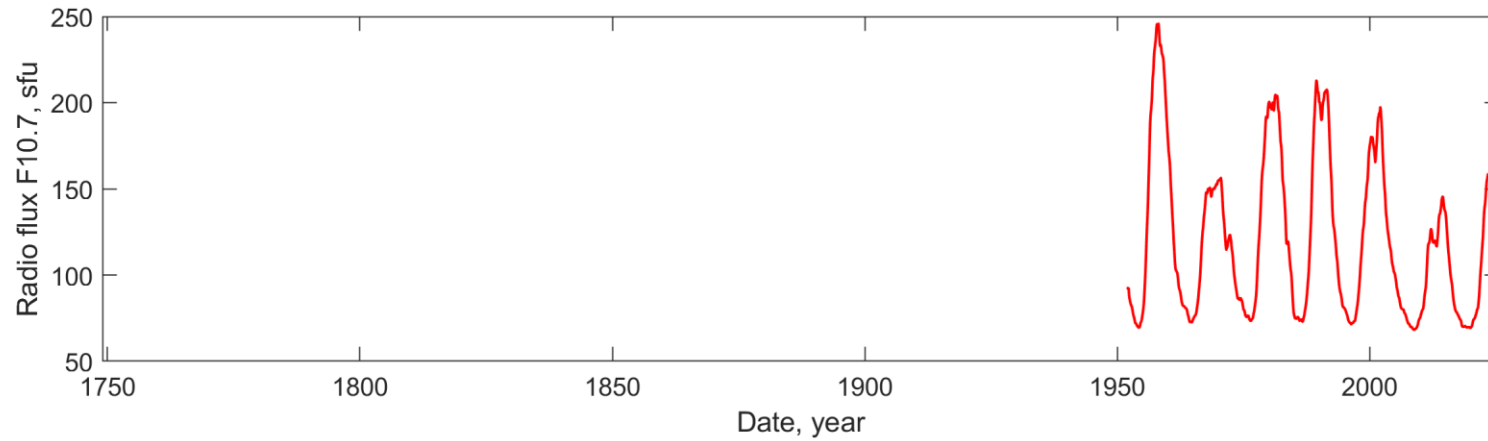
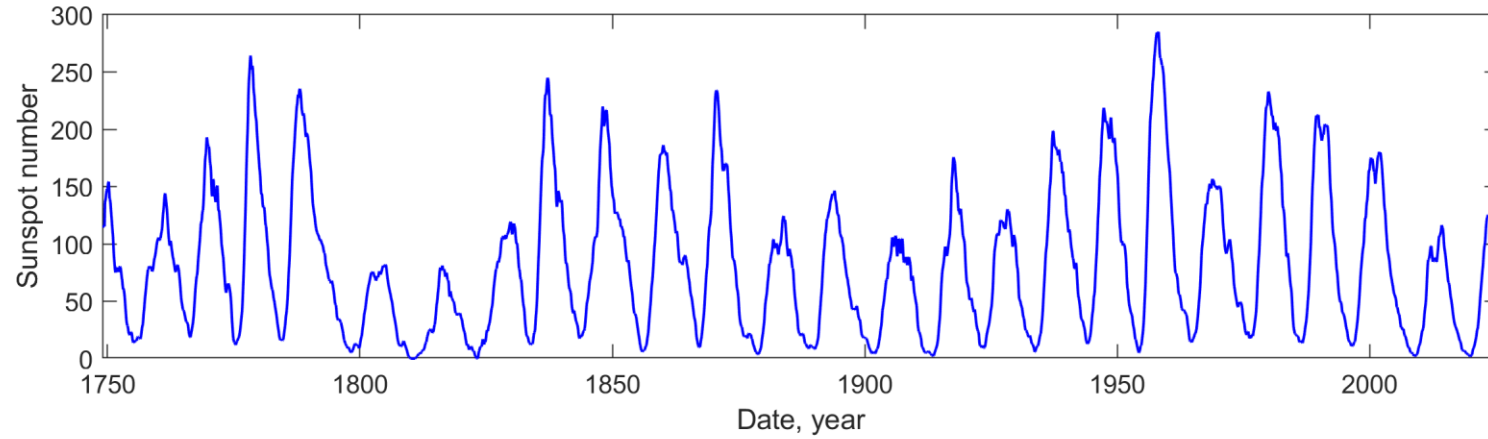
13-month sequent monthly mean sunspot numbers



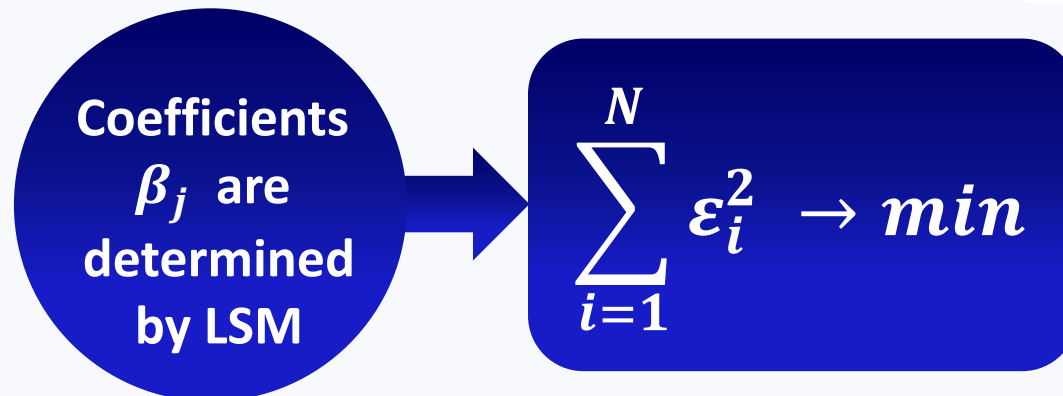
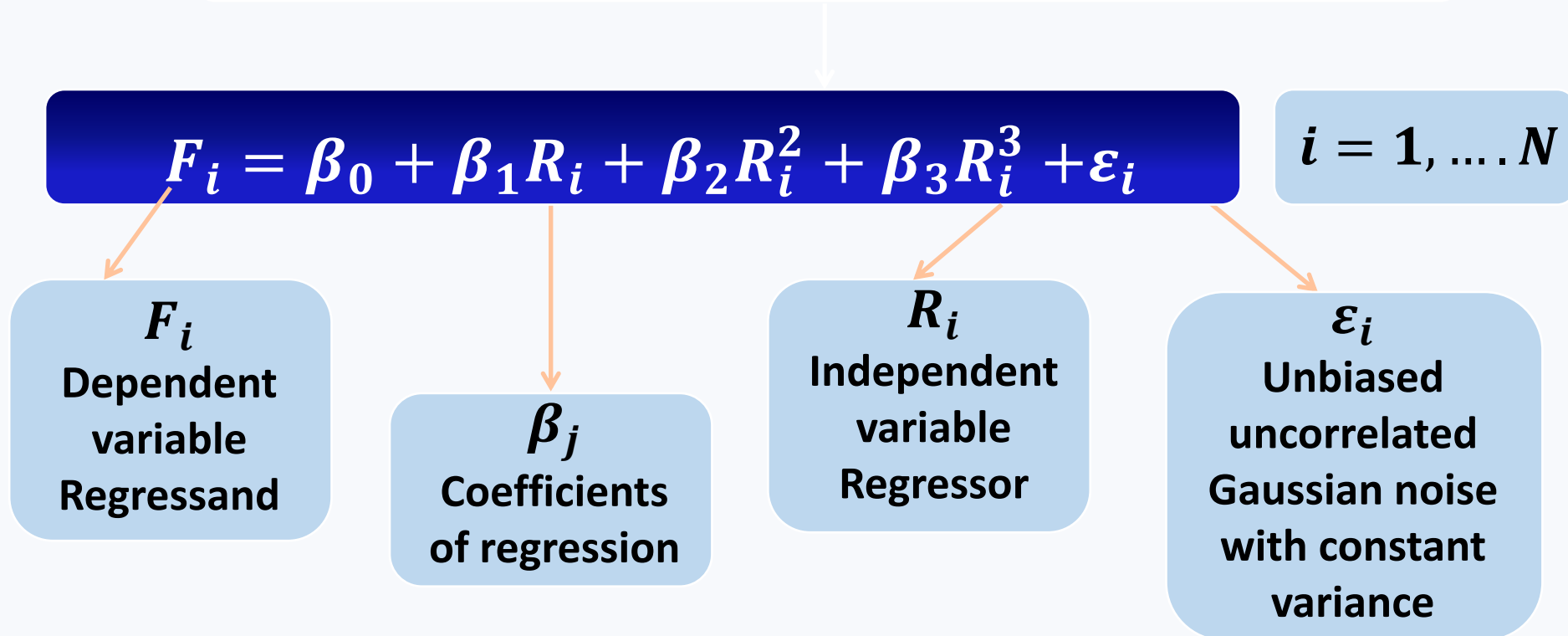
13-month running mean $\bar{R}$

$$\frac{1}{24}R_{i-6} + \frac{1}{12}(R_{i-5} + R_{i-4} + \cdots + R_{i-1} + R_i + R_{i+1} + \cdots + R_{i+5}) + \frac{1}{24}R_{i+6}$$

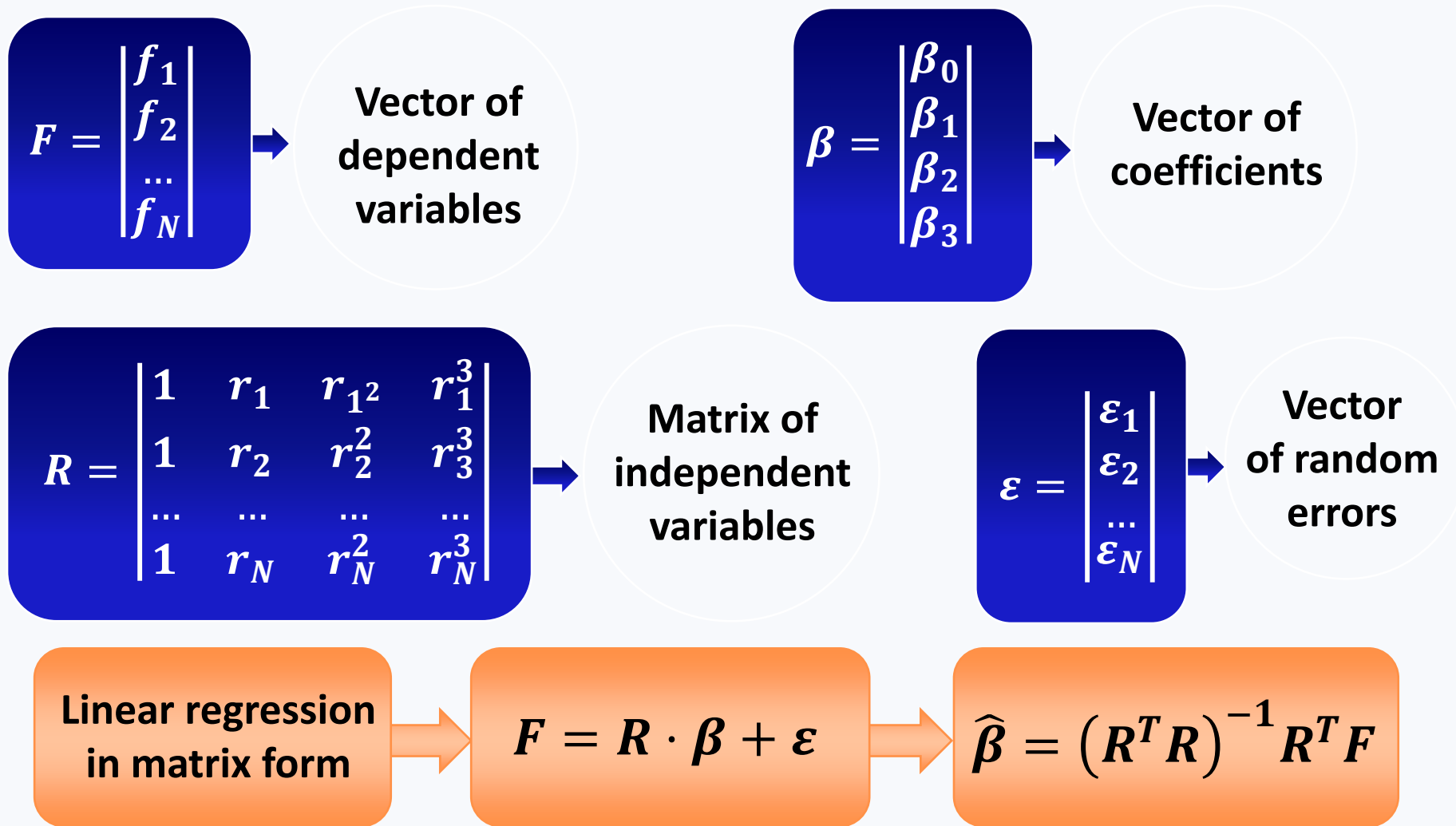
Smoothing: 13-month running mean



Multi-dimensional linear regression



Multi-dimensional linear regression

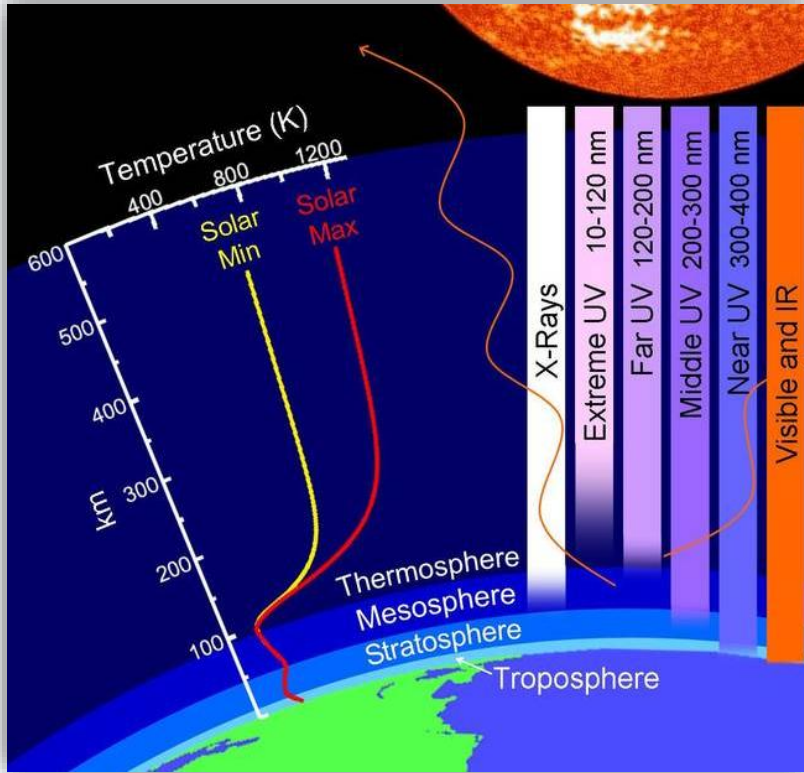


Estimation error of solar radio flux F10.7

Covariance
matrix of
estimation
error



$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (f_i - \hat{f}_i)^2$$



Properties of Earth's atmosphere, NASA

Solar EUV irradiance



Variability in the thermosphere density and temperature



Atmospheric drag change



Decay and following re-entry



Estimated via
atmosphere models



Require **F10.7** and **F30**
as a proxy for **solar input**



What will this assignment give you?

A practical path from noisy historical observations to reliable reconstruction and prediction.

1 Observe

Work with two solar activity indicators: sunspot number and F10.7 radio flux.

2 Smooth

Reduce short-term variability to reveal the relationship between the indicators.

3 Model

Build a regression model and determine its coefficients by LSM.

4 Reconstruct

Extend F10.7 backward in time and add uncertainty bounds.

From data to model: $F_i = \beta_0 + \beta_1 R_i + \beta_2 R_i^2 + \beta_3 R_i^3 + \epsilon_i \rightarrow$ uncertainty: $\pm 3\sigma$

Why it matters

F10.7 is used as a solar input proxy in atmosphere models, which support orbit specification, re-entry services, collision avoidance, and space-debris modelling.

You do not just fit a curve — you learn how to make historical data useful for prediction and decisions.